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ABSTRACT

Here goes the abstract

1. Introduction

In recent years there has been a global expansion of renewable power generation Shahbaz, Siddiqui, Siddiqui, Jiao and Kautish (2023); Hassan, Algburi, Sameen, Al-Musawi, Al-Jiboory, Salman, Ali and Jaszczur (2024), driven by de-carbonization efforts and declining costs in renewable technologies Jäger-Waldau (2025); Joshi, Rao and Banerjee (2025); Manservigi, Castorino, Spina, Venturini and Gonzalez-Salazar (2026). This rapid deployment of renewable generation has been mostly due to additions in solar (at both utility and residential scale), hydro and wind capacity Shahbaz et al. (2023); Hassan et al. (2024). Currently, annual renewable capacity additions outpace conventional capacity additions IRENA (2025), and this upward trend will likely continue in the forecoming decades Agency (2019); Raturi (2021).

However, the rapid expansion of renewable generation is leading to grid integration difficulties due to inherent variability and uncertainty in the output Li, Shi, Cao, Wang, Kuang, Tan and Wei (2015); Liu and He (2023). Spatiotemporal mismatches between power supply and demand may arise if capacity planning and grid infrastructure development do not keep pace Ming, Kun and Jun (2013); Bird, Lew, Milligan, Carlini, Estanqueiro, Flynn, Gomez-Lazaro, Holttinen, Menemenlis, Orths et al. (2016); O'Shaughnessy, Cruce and Xu (2020); Li, Wang, Guo, Xin, Zhang, Liu and Feng (2024). Oversupply of renewable generation — occurring during periods of low demand or high wind and solar resource availability — can result in curtailment of a fraction of the generated energy Denholm, O'Connell, Brinkman and Jorgenson (2015); Bird et al. (2016); O'Shaughnessy et al. (2020). This issue is further compounded by the increasing penetration of residential-scale distributed generation, which is generally excluded from centralized dispatch scheduling owing to its negligible individual capacity Li et al. (2015); Bhattarai, Maraseni and Apan (2022). As a result, these challenges place increasing demands on power system flexibility and energy storage capacity, both of which are essential for balancing supply and demand in high-penetration renewable energy systems Denholm and Mai (2019); **?**); Nejla, Saidani and Besbes (2025).

In particular, South America has also experienced accelerated penetration of renewable generation in recent years, with wind and solar photovoltaic capacity accounting for more than half of annual capacity additions International Energy Agency (2023). Although renewable penetration has been uneven across countries, several are already experiencing increasing curtailment rates von Papp, Moraga and Henríquez (2022); Silva, Renolphi, Vieira, Lourenço, Salles and Monaro (2025). Chile has recorded curtailment rates of up to 20% of total annual renewable generation, driven mainly by a geographical mismatch between solar generation — concentrated in the northern Atacama region — and demand centers located farther south von Papp et al. (2022); Silva et al. (2025). In Brazil, by contrast, the rapid expansion of distributed generation has reshaped the load profile, also resulting in high curtailment rates da Ponte, Rego and Júnior (2025); Silva, Vieira, Rego, Simone, Lourenço and Salles (2026). Despite these challenges, the region's abundant hydroelectric capacity Farfan and Breyer (2017); International Energy Agency (2023) represents an additional advantage for renewable energy integration, offering flexible storage and dispatch capabilities that can help mitigate variability and curtailment Wan and Parsons (1993); Hingray, Favre, Sauquet and Anquetin (2014).

Bolivia's current power mix is dominated by hydro and thermal generation—accounting for approximately 70% and 20% of installed capacity, respectively — with only a small share of non-hydro renewables Autoridad de Fiscalización de Electricidad y Tecnología Nuclear (AETN) (2026) —. Efforts are underway to further decarbonize the electricity sector: national development plans Viceministerio de Electricidad y Energías Renovables (2025) and Bolivia's Nationally Determined Contributions (NDC) Ministerio de Medio Ambiente y Agua and Autoridad Plurinacional de la Madre Tierra (2021) foresee substantial growth in centralized renewable capacity. The accelerated uptake of distributed solar PV observed in the region Chueca, Weiss, Celaya, Ravillard, Ortega, Tolmasquim and Hallack (2023) suggests that distributed generation growth represents a plausible complementary pathway.

Given this growing complexity in power systems, energy modeling has become a crucial tool for exploring

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supply and demand scenarios that support long-term policy decisions Huntington, Weyant and Sweeney (1982); Pfenninger, Hawkes and Keirstead (2014). However, few studies have modeled the impacts of renewable energy penetration on power systems in South America. In Bolivia, Fernandez-Vazquez et al. Fernandez Vazquez, Vansighen, Fernandez Fuentes and Quoilin (2022) developed a long-term optimization model of the Bolivian energy sector using OSeMOSYS, based on a trend-based projection of power demand, and analyzed policy-based scenarios to induce decarbonization of the power system. However, the specific impacts of renewable capacity expansion — particularly distributed generation — on transmission requirements, thermal fuel consumption, greenhouse gas (GHG) emissions, and energy curtailment remain unexplored. This study aims to analyze the impacts of exogenous renewable capacity additions — both centralized and distributed — on power transmission, fuel consumption in thermal power facilities, GHG emissions, and energy curtailment. An intraday hydroelectric dispatch strategy is also proposed to mitigate renewable energy curtailment. For this purpose, we constructed a regionalized model of the Bolivian energy sector using the LEAP and NEMO modeling tools.

The paper is structured as follows: Section 2 describes the particularities of Bolivia's energy and power system, the modeling approach implemented in this study, the characterization of energy and electricity demand, and the current power generation fleet along with the scenarios considered. Section 3 presents the results of the scenario analysis and comparison in terms of their effects on overall power demand and supply, fuel savings, GHG emissions, regional electricity balances and flows, and total costs within the NIS.

2. Methodology

2.1. Case study

Electricity demand currently represents approximately 11% of Bolivia's overall energy demand and has grown at an annual rate of approximately 5.7% in recent years Ministerio de Hidrocarburos y Energías, Viceministerio de Planificación y Desarrollo Energético (2023, 2024). This demand is predominantly met by natural gas (NG) and hydroelectric generation. By the end of 2024, the installed capacity of the National Interconnected System (NIS) totaled 3580 MW, of which 2573 MW corresponded to thermoelectric sources and 758 MW to hydroelectric sources, with the remaining 249 MW distributed across wind, solar, and biomass. Approximately 98% of thermoelectric capacity is NG-based, with the remainder corresponding to diesel engines; the NG-based fleet, in turn, is composed of 66% combined-cycle, 30% turbo-gas, and 4% turbo-vapor units.

NG-based generation accounts for most of Bolivia's capacity expansion over the last three decades Ballivian Cabrera (2024). More recently, however, a steep decline in domestic NG production Ministerio de Hidrocarburos y Energías, Viceministerio de Planificación y Desarrollo Energético (2024) has motivated the development of national plans and policies aimed at decarbonizing the power sector Balderrama, Broad, Sevillano, Alejo and Howells (2017); Fernandez Vazquez et al. (2022). The capacity expansion plan assessed in this study, prepared by the Ministry of Energy, includes two ambitious scenarios of 100% renewable capacity additions up to 2037.

Table 1
Power lines

Line	Nodes	Capacity [MW]
L1	Centro-Norte	390
L2	Centro-Sur	260
L3	Centro-Oriente	165
L4	Oriente-Norte	35
L5	Centro-Oriente	500

Bolivia's NIS has no current linkages with the national grids of bordering countries and can therefore be considered an isolated system; it currently covers eight out of nine departments. Given this configuration, Bolivia's energy system was represented in this study as a multi-nodal system, consisting of five nodes or regions obtained by grouping the administrative divisions as follows:

- Pando
- Norte (La Paz + Beni)
- Centro (Cochabamba + Oruro)
- Oriente (Santa Cruz)
- Sur (Tarija, Potosí, and Chuquisaca)

This regionalization was based on prior energy modeling studies conducted in Bolivia that proposed this specific departmental aggregation (see Huallpara, Navia Orellana, Gomand, Balderrama, Pavičević and Quoilin (2023)). Accordingly, this structure was replicated in the present study to facilitate comparison with independently conducted modeling exercises.

Figure 1 shows the location of the nodes and the representation of the NIS, for which two power lines connecting Centro and Oriente, one line connecting Centro and Sur, one line connecting Centro and Norte, and one connecting Oriente and Norte were considered. Pando Region is not currently interconnected with the other regions. AGREGAR LOSSES The main features of the power lines are detailed in Table 1.

2.2. Modelling approach

In this study, the Low Emissions Analysis Platform (LEAP) and the Next Energy Modeling System for Optimization (NEMO) were used to model Bolivia's energy sector. LEAP is a modeling tool developed by the Stockholm Environment Institute Heaps (2022). It has been widely used by researchers and organizations to project future energy demand and supply and greenhouse gas (GHG) emissions, and to assess energy policies and scenarios Emodi, Emodi,

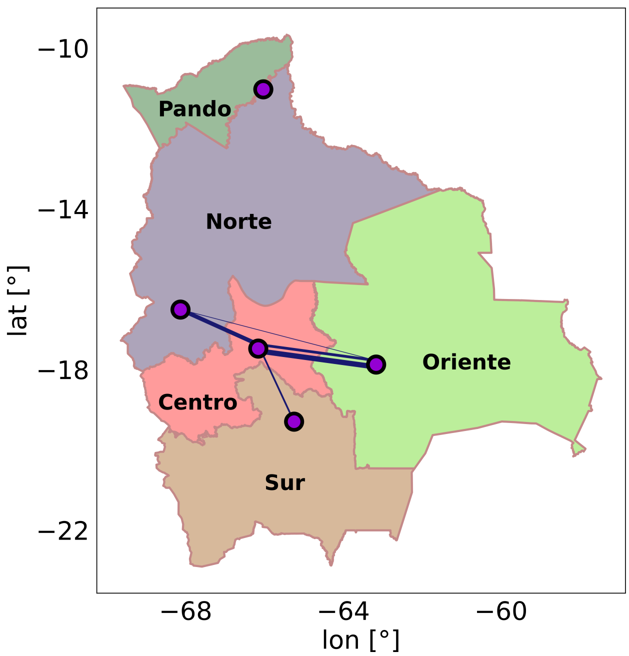

Figure 1: Regional nodes included in the model and power lines

Murthy and Emodi (2017); Shahid, Ullah, Imran, Mahmood and Arentsen (2021); Handayani, Overland, Suryadi and Vakulchuk (2023). It supports the implementation of multiple modeling approaches within a single model, including bottom-up Fleiter, Worrell and Eichhammer (2011) and top-down methodologies Rogan, Cahill, Daly, Dineen, Deane, Heaps, Welsch, Howells, Bazilian and Ó Gallachóir (2014), and it allows the formulation and comparison of different energy scenarios. NEMO is an open-source tool for energy system optimization Stockholm Environment Institute (2026). Combined with LEAP, it enables additional modeling capabilities, such as least-cost optimization of energy supply and demand, multi-regional trade simulation, nodal network configuration, and energy storage modeling. In this study, we implemented a bottom-up approach and enabled NEMO's optimization module, together with the Cbc solver, for non-renewable capacity expansion, nodal simulation, and electricity dispatch.

The structure of both energy demand and supply — including regional features — was replicated in LEAP using information from national energy balances and inventories Ministerio de Hidrocarburos y Energías, Viceministerio de Planificación y Desarrollo Energético (2023, 2024); Autoridad de Fiscalización de Electricidad y Tecnología Nuclear (AETN) (2026), which provide annual energy demand volumes disaggregated by sector (industry, residential, transport, etc.) and by energy source (electricity, gasoline, diesel, etc.), as well as an updated generation fleet inventory. The base year of the simulation was set to 2023, and the simulation covers the 2023–2050 period.

Both power load curves as well as maximum availability curves for hydroelectric, solar and wind plants were configured with two-hourly resolution grouped in a wet season (november to april) amd a dry season (may to october), resulting in XXX tome slices per year.

2.3. Demand

As mentioned before, energy demand was modeled for 5 geographical regions (Pando, Norte, Centro Oriente, Sur) and for 7 sectors: residential, transport, industry, commercial + public, agriculture + fishery + mining, and construction. For the residential sector, the number of households represents the unit of consumption. We used population projections from the national bureau of statistics (INE in spanish Instituto Nacional de Estadística de Bolivia (2024)) to calculate the projected number of households. Households were then disaggregated into urban and rural subsectors, and rural households were further divided into electrified and non-electrified. Final energy uses — such as cooking, water heating, food conservation, and lighting — were then characterized in terms of coverage, efficiency, fuel share, and energy intensity (DE DONDE SALIO ESTO, CITAR). Miscellaneous thermal and electrical uses were also characterized. All electrical appliances were associated with a load curve (EN QUE TRABAJO PREVIO SE CARACTERIZO ESTO).

The transport sector was sub-divided into: road (light-duty, medium-duty, and heavy-duty), rail, air, waterway, and cable car. The vehicle fleet was regionalized according to inventories provided by INE.

The industrial sector was disaggregated into large-scale and small-scale units. For small-scale units, previous sectoral surveys (CITA) allowed further disaggregation into seven categories: food, beverages and tobacco; non-metallic minerals; paper, wood and furniture; machinery and metals; textiles; clothing and leather; and chemicals, plastics and pharmaceuticals. Large-scale units were classified into seven categories according to a previous sectoral survey from INE: food and beverage; cement; bricks; metals; paper, cardboard and wood; chemicals and petrochemicals; and construction materials. For this disaggregation level, we included the consumption share of different fuel types for the years 2021–2023. The fuels considered are electricity, diesel, gasoline, jet fuel, biomass, natural gas (NG), and liquefied petroleum gas (LPG).

Only one energy demand growth scenario was considered in this modeling exercise. For the transport; industry; commercial and public; agriculture, fishery and mining; and construction sectors, gross domestic product (GDP) was taken as the driver of energy demand. Sub-national reports from INE allowed the regional and sectoral disaggregation of GDP. An annual growth rate of 2.8%, based on historical trends, was considered for the simulation period. For the residential sector, the number of households was taken as the driver of energy demand. INE projects an overall population growth of 20%, reaching a total of 13.4 million inhabitants by 2050. Data from the last census allowed the regional and

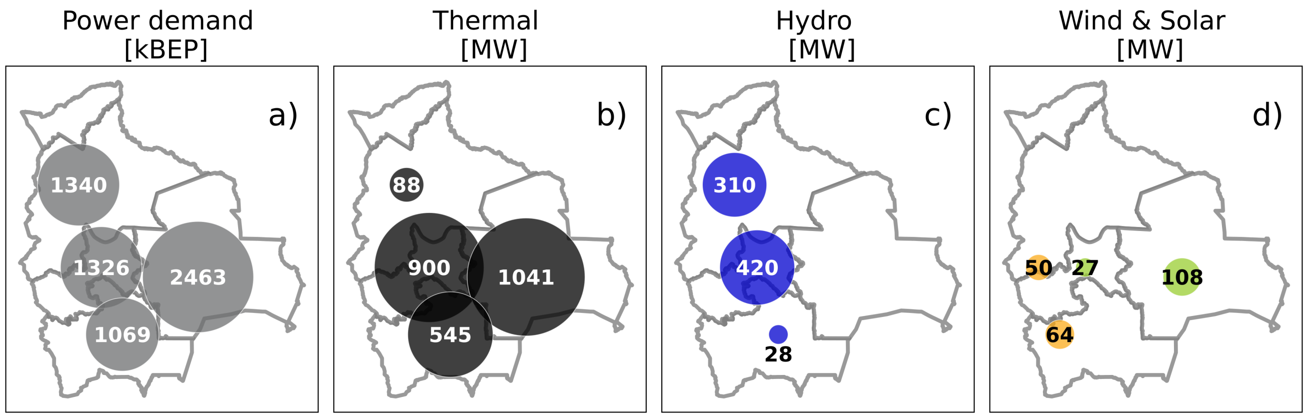

Figure 2: Current (2023) annual electricity demand (a); thermal (b), hydro (c) and wind/solar (d) installed capacity

urban/rural disaggregation of population. For each region, both total population and number of households were extrapolated from the base year onward, assuming that the regional proportion remains constant — an assumption that held for the 2012–2024 period. In order to project the regional urban/rural share of households, the ratio of urban-to-rural growth rates was computed. Then, an objective function was included in the projection so that the overall projected national number of urban households matched CEPAL's projection.

An analysis of the current energy demand structure shows that electricity represents approximately 11.8% of the overall energy demand. Electricity demand within the NIS is concentrated in the Oriente region (39%), followed by Centro, Norte (22% each), and Sur (17%) (Figure 2, panel a). The most electricity-consuming sector is residential (37%), followed by industry (24%); commercial and public (23%); agriculture, fishery and mining (11%); construction (3%); and transport (0.4%).

2.4. Power supply

As mentioned above, Bolivia's current generation mix is predominantly thermal, with hydropower as a significant secondary source. Around 70% of installed capacity is fossil-fueled, concentrated mainly in the Oriente, Centro, and Sur regions (figure 2b)), while hydro capacity is located in the Centro and Norte regions (figure 2c)). Wind and solar capacity, which together account for less than 1% of installed capacity, is concentrated in the Oriente and Centro/Sur regions (figure 2d)). Against this backdrop, this modeling exercise aims to assess the impact of expanding both centralized and distributed renewable capacity in the NIS. To this end, two centralized capacity expansion scenarios were built on top of the demand growth base scenario: Business as Usual (BAU) and Alternative (ALT). Both scenarios represent the expansion of centralized renewable capacity (wind, hydro, and solar) following the official capacity expansion schedule for the 2024-2050 period.

The BAU scenario contemplates a short-term capacity expansion that includes a 261 MW addition of wind capacity in the Oriente region (Figures 3 and 4). It proposes the development of two hydroelectric projects: Ivirizú (291 MW), located in the Central region and comprising two hydroelectric plants (Sehuencas and Juntas); and Miguillas (210 MW) in the Norte region, comprising two cascade hydroelectric plants, Umapalca (86 MW) and Palillada (119 MW). It also includes the addition of 340.5 MW of solar capacity in the Centro and Norte regions. All these capacity additions are configured in LEAP as exogenous capacity additions. Finally, the NEMO optimization assigns a total endogenous capacity of 160 MW of turbo-gas (TG) units for the simulation period. The ALT scenario contemplates a medium-term capacity expansion that adds an additional 631 MW, 1525 MW, and 1220 MW of wind, hydro, and solar capacity relative to the BAU scenario. Wind capacity additions are concentrated in the Oriente (90%) and Centro (10%) regions, and are distributed in units ranging from 25 to 100 MW. Hydro capacity additions are associated with the exploitation of the Río Grande basin, and comprise the Rositas (600 MW), La Pesca (460 MW), Cañahuecal (380 MW), and Okitas (85 MW) facilities across the Centro and Oriente regions.

All centralized capacity additions were configured in LEAP as exogenous capacity additions. Maximum availability of wind, solar, and hydro was represented by availability curves derived from historical data that display daily fluctuations. A minimum reserve margin of 30% was assigned for the NIS, and differential NG production costs across regions were configured to represent the regional availability of this resource. NEMO optimization allowed for endogenous capacity expansion when needed to meet the reserve margin requirement.

Both scenarios (BAU and ALT) were then combined with two sub-scenarios representing different Solar Distributed (SD) capacity trajectories, resulting in four scenarios in total. BAU and ALT incorporate a total of 100 MW of SD capacity by 2050, whereas BAU-SD and ALT-SD incorporate a total of 1,000 MW by 2050. SD capacity was added

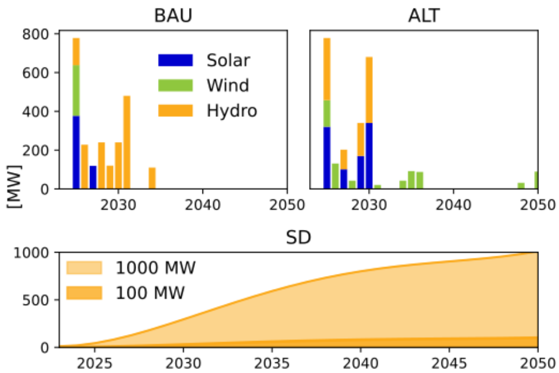

Figure 3: Centralized generation capacity additions for BAU and ALT scenarios (top row), and for SD generation (below)

following logistic growth curves Joshi et al. (2025) (figure 3). SD capacity was regionally disaggregated according to current regional shares of power demand and incorporated into LEAP as a negative electricity demand source, thereby bypassing transmission and distribution losses.

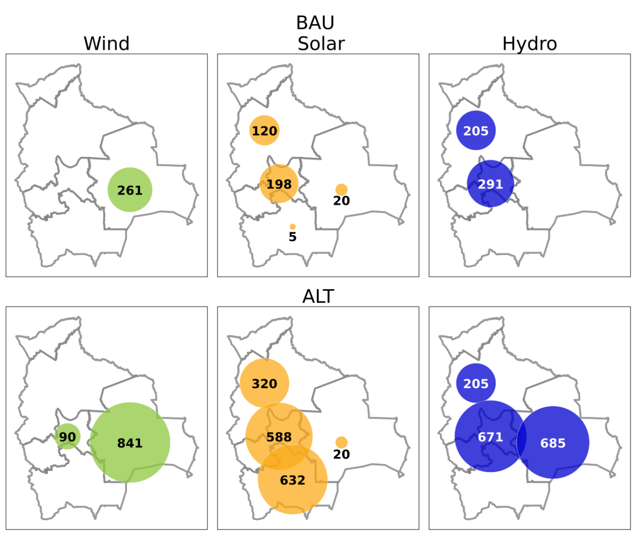

Figure 4: Regional centralized capacity additions [MW] according to BAU and ALT scenarios

For each region, we analyzed the evolution of primary and secondary energy sources within the power mix, as well as hourly power flows between regions. A regional balance of electricity generation by source and net trade was also compiled for 2050. GHG emissions were compared across scenarios using 100-year global warming potential (GWP) factors and the built-in Technology and Environmental Database, which contains emission profiles for different generation technologies from the Intergovernmental Panel on Climate Change (IPCC), the US Department of Energy, and the International Energy Agency.

Finally, a total cost analysis of the power system was performed, along with a comparison between scenarios. Total costs include capital and Operation and Maintenance (OM) costs for different generation technologies (see Table XX),

Table 2
Capital and O&M costs for different power generation technologies

Technology	Capital [USD/MW]	O&M [USD/MW-yr]
TG	855	8
TV	1,800	20
Diesel	1,200	15
CC	1,400	12
Solar	1,000	22
Solar (new)	850–940	20–22
SD	1,200	20–22
Wind	1,500	36–39
Wind (new)	1,300–1,450	39
Hydro	1,500	7
Rositas (hydro)	3,350	38.8
Cachuela (hydro)	1,880	40
Rio Grande (hydro)	3,446	39.5
Other hydro	2,029	29.6

along with differential costs associated with displacing NG-based generation and SD installation costs. Capital and OM costs are incorporated into the analysis through an annual payment function that amortizes investments over time. This function accounts for equipment lifetime and the interest rate — or systemic discount rate — which was set at 12% for the entire model. Installation and OM costs for new wind and solar capacity are assumed to decline over time **?**.

Regarding the opportunity cost of NG — the main input for fossil-fueled generation — non-utilized NG was assumed to be available for export at a price of 5 US$/million BTU. A price of 120 US$/MWh was assigned to diesel-based generation. The installation cost of SD photovoltaic panels was set at 1200 US$/kW, comprising total capital cost and distribution adequacy costs (i.e., costs associated with reinforcing distribution infrastructure to accommodate distributed generation). Capital costs are assumed to decrease over time, while distribution adequacy costs increase, resulting in an overall installation cost that declines to 584 US$/kW by the end of the simulation period. All values were calculated at constant 2025 prices, using the 12% discount rate described above.

3. Results

3.1. Demand

The addition of SD (distributed generation) reduces the net power demand observed in the NIS. Comparison between scenarios (BAU-SD/ALT-SD vs. BAU/ALT) shows that an additional capacity of 900 MW of SD reduces net demand by about 9% by 2050. The net accumulated energy savings over the simulation period amount to approximately 22,000 and 25,000 GWh, respectively, representing nearly twice the total annual demand of the base year.

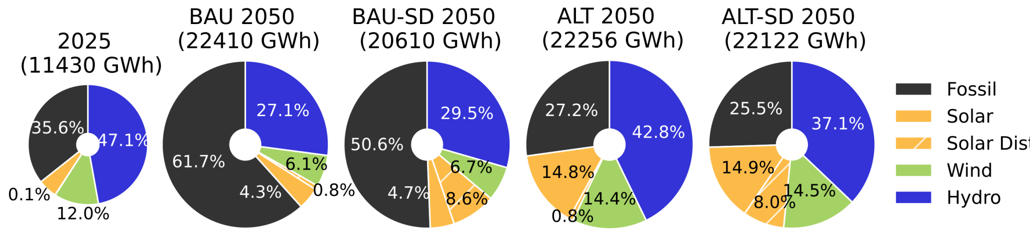

Figure 5: Current (2023) and projected annual electricity generation according to BAU, BAU-SD, ALT and ALT-SD scenarios

3.2. Power supply

During the simulation period, NEMO assigned a total endogenous capacity of 160 MW of turbo-gas (TG) units under the BAU scenario, whereas no endogenous capacity expansion was observed under ALT.

Regarding power generation, the addition of SD capacity coincides with a decrease in thermal generation in both the BAU and BAU-SD scenarios. In the first scenario, SD generation represents less than 1% of annual generation by 2050, while thermal generation escalates from 36% in the base year to 61% (figure 5). In the second scenario, by contrast, SD generation represents nearly 9% of annual generation by 2050, while thermal generation only escalates to 51%. For the ALT and ALT-SD scenarios, there is a marked reduction in thermal generation, reaching 27% and 26% of annual generation by 2050, respectively, though this reduction is mostly associated with the massive deployment of centralized renewable capacity. The addition of 900 MW of SD (comparing ALT and ALT-SD) coincides with a reduction in hydro generation (42% of annual generation in ALT vs. 37% in ALT-SD).

The addition of SD capacity produces positive externalities in terms of fuel savings and GHG emissions. An additional 900 MW of SD capacity produces greater fuel savings for the BAU scenario than for the ALT scenario. Fuel savings for the BAU-SD scenario reach 960 kBEP annually by 2050 compared to BAU, with accumulated savings of 15,000 kBEP over the simulation period — nearly the total inputs for power generation in the base year. For the ALT-SD scenario, by contrast, accumulated savings total only 1,930 kBEP compared to ALT. GHG emissions follow a similar pattern: accumulated emission savings are greater for the BAU-SD scenario compared to BAU (9,900,000 metric tonnes of CO_2-equivalent, Figure 6) than for the ALT-SD scenario compared to ALT (1,170,000 metric tonnes of CO_2-equivalent).

Overall costs related to the NIS present differences across scenarios. Costs are higher for the ALT scenario (1,600 million USD) than for the BAU scenario. Fuel-related savings of around 1,000 million USD are exceeded by higher capital costs (see Table 3). The addition of SD capacity produces modest increases in overall costs: 185 and 206 million USD for BAU-SD and ALT-SD compared with BAU and ALT, respectively. Savings related to fuel replacement are higher for BAU-SD, since SD generation replaces thermal generation. In ALT-SD, by contrast, SD generation displaces a higher proportion of centralized renewable generation.

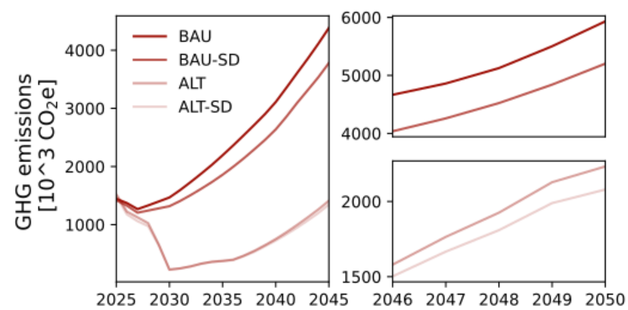

Figure 6: Annual GHG emmisions according to BAU, BAU-SD, ALT and ALT-SD scenarios

Table 3
Costs (BAU ALT scenarios) and differential costs (BAU-SD ALT-SD scenarios). Values are expressed in MM USD

	BAU	(BAU-SD) -BAU	ALT	(ALT-SD) -ALT
Fuel	2,227	-171	1,192	-25
Capital + O&M	6,963	-1	11,605	-128
SD panels	39	360	39	360
Total	9,230	185	12,837	206

3.2.1. Inter regional power flows

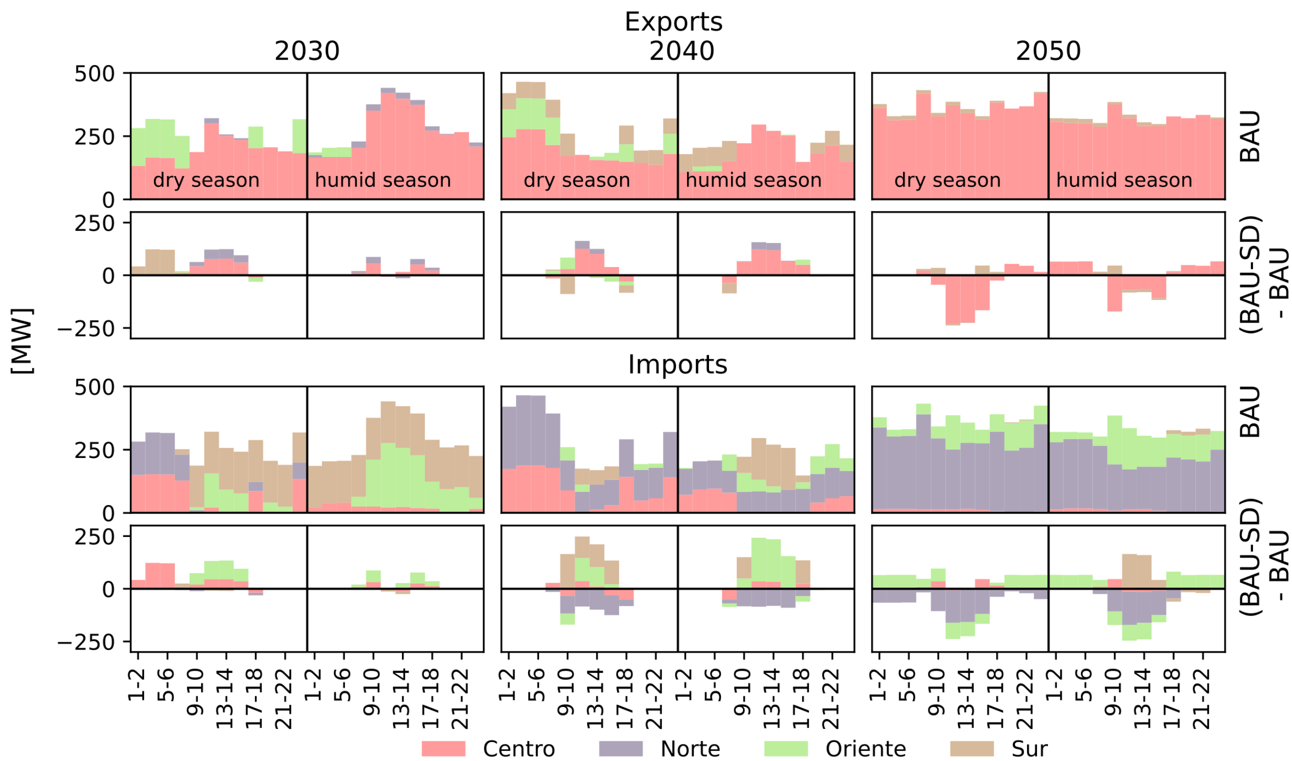

Figure 7: Current (2023) and projected annual electricity generation according to BAU, BAU-SD, ALT and ALT-SD scenarios

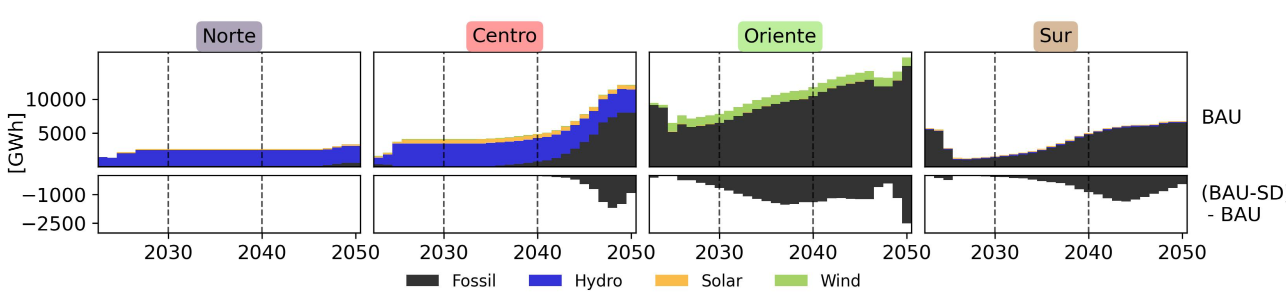

Figure 8: Current (2023) and projected annual electricity generation according to BAU, BAU-SD, ALT and ALT-SD scenarios

Figure 9:

Table 4

4.

CRediT authorship contribution statement

Emilio Bianchi: Formal analysis, Investigation, Methodology, Visualization, Writing 1–original draft. **Ignacio Sagardoy:** Conceptualization, Formal analysis, Investigation, Methodology, Validation. **Francisco Lallana:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Validation. **Gustavo Nadal:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision. **Gonzalo Bravo:** Funding acquisition, Project administration. **Gonzalo Bravo:** Conceptualization of this study, Methodology, Software.

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